

1. $z = x \sin y$ 求 $\frac{\partial z}{\partial y}$
 $x \cos y$

2. $\lim_{(x,y) \rightarrow (1,0)} \frac{\sin(xy)}{xy} =$
 $\downarrow$
 $\frac{xy}{xy} = 1$

3. $\lim_{(x,y) \rightarrow (0,0)} \frac{2 - \sqrt{xy+k}}{xy}$
 交换 xy 不变.

本题 $\lim_{(x,y) \rightarrow (0,0)} \lim_{z \rightarrow 0} \frac{2 - \sqrt{z+k}}{z}$
 $z = xy$

4. $y'' - 2y' - 3y = 0$ 通解.
 $r^2 - 2r - 3 = 0$
 $(r-3)(r+1) = 0$
 $y = C_1 e^{3x} + C_2 e^{-x}$

5. $\vec{u} = (\sqrt{3}, 1, 0)$ $\vec{v} = (1, \sqrt{3}, 0)$ 夹角.
 $\cos \theta = \frac{2\sqrt{3}}{4} = \frac{\sqrt{3}}{2}$
 $\theta = 30^\circ$

6. $z = 2x^2 + 3y^2$ 在点 $(1, 2)$ 梯度.
 $z_x = 4x$ $z_y = 6y$
 $(4, 12)$

13. $\sum_{n=1}^{\infty} x^n$ 收敛半径.

$$\lim_{n \rightarrow \infty} \frac{x^{n+1}}{x^n} = x < 1 \quad R=1$$

14. $z = f(x, x+y)$ 求 dz .

15. 知 α, β, γ 二类曲面转一类

$$\int \int_{\Sigma} P dx dy + Q dy dz + R dz dx$$

$$\int \int_{\Sigma} (P \cos \gamma + Q \cos \alpha + R \cos \beta) ds.$$

题目中他出现了 PQR 对应 $dx dy$ 的更换
注意这点.

正确应该是 $P dy dz + Q dz dx + R dx dy$ 这样.

16. 大题始.

$$2x^3 + 2y^3 + 3x^2 - 3y^2 - 12x - 12y + 1 \quad \text{极小与极大.}$$

$$17. \text{求 } I = \int_{\gamma} xy ds \quad D \text{ 由 } y^2 = x \text{ 与 } y = x^2 \text{ 围成}$$

~~48~~ 45/8

18. 常数变易法求 $y' + y = e^{-x}$ 通解.
限定这个条件.

建议可以先用公式法算结果 回忆常数变易法求过程

7. $f = xy^2 + y^2z + zx$ 求 f_{xx}
 $f_x = y^2 + 2zx$
 $f_{xx} = 2z$

8. 过点 $(2, 1, 3)$ 与 $3x + 2y - z = 0$ 垂直直线方程.
 $\vec{s} = (3, 2, -1)$
 $\frac{x-2}{3} = \frac{y-1}{2} = \frac{z-3}{-1}$

9. 隐函数求导 相当重点.
 $x^2 + y^2 + z^2 - 4z = 0$ 求 $\frac{\partial z}{\partial y}$.

$$\frac{\partial z}{\partial y} = - \frac{F_y}{F_z} = \frac{y}{z-2}$$

$$F_y = 2y$$

$$F_z = 2z - 4$$

10. $\sum_{n=1}^{\infty} (\frac{3}{2})^n$ 敛散性.

$n \rightarrow \infty \quad \lim (\frac{3}{2})^n \neq 0$. 发散.

11. 交换积分顺序 $\int_0^1 dx \int_0^{\sqrt{x}} f(x, y) dy$.

12. 格林.

$$\oint_L y^2 dx + x^2 dy = \iint_D (2y - 2x) dx dy.$$

19. $f(x) = e^{2x}$ (1) 求 n 阶导
(2) 将 $f(x)$ 展开成 x 的幂级数展开.

关于(2) 十分基础, 看过宋浩应该都懂.

20. a, b, c 皆为正数.

S 是方向向上半椭球面

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1 \quad (z \geq 0)$$

$$I = \oint_S$$

1. $z = x \sin y$ 求 $\frac{\partial z}{\partial y}$

$x \cos y$

2. $\lim_{(x,y) \rightarrow (1,0)} \frac{\sin(xy)}{xy} = 1$

$= \frac{xy}{xy} \rightarrow$

3. $\lim_{(x,y) \rightarrow (0,0)} \frac{2\sqrt{xy+4}}{xy}$

~~交换~~ 交换 xy
发现不变

$\lim_{(x,y) \rightarrow (0,0)} \frac{2 - \sqrt{z+4}}{z}$
 $z = xy$
 $\lim_{z \rightarrow (0,0)}$

$\lim_{z \rightarrow 0} \frac{-\frac{1}{2\sqrt{z+4}}}{1} = -\frac{1}{16}$

4. $y'' - 2y' - 3y = 0$ 通解

$r^2 - 2r - 3 = 0$

$(r-3)(r+1) = 0$

$y = C_1 e^{3x} + C_2 e^{-x}$

5. $\vec{u} = \sqrt{3}, 1, 0, \vec{v} = 1, \sqrt{3}, 0$ 夹角

$\frac{d\vec{v}}{2 \cdot 2} = \frac{2\sqrt{3}}{4} = \frac{\sqrt{3}}{2}$

$\theta = 30^\circ$

6. $z = 2x^2 + 3y^2$ 在点 $(1, 2)$ 梯度

$z_x = 4x \quad z_y = 6y$

$(4, 12)$

7. $f = xy^2 + y^2z + zx^2$

$f_x = y^2 + 2zx$

$f_{xx} = 2z$

8. 过点 $(2, 1, 1)$ 与 $3x + 2y + z = 0$ 垂直

$\vec{z} = (3, 2, -1)$

$\frac{x-2}{3} = \frac{y-1}{2} = \frac{z-1}{-1}$

9. 隐函数求导 相当重点

$x^2 + y^2 + z^2 - 4z = 0$ 求 $\frac{\partial z}{\partial y}$

$\frac{\partial z}{\partial y} = -\frac{F_y}{F_z}$

$F_y = 2y$

$F_z = 2z - 4$

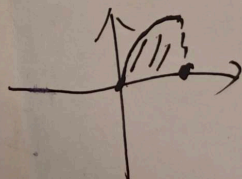
$\frac{y}{z-2}$

10. $\sum_{n=1}^{\infty} \left(\frac{z}{2}\right)^n$ 敛散性

$\left|\frac{z}{2}\right|^n = \frac{1}{\left(\frac{2}{z}\right)^n}$

$\lim_{n \rightarrow \infty} \left(\frac{z}{2}\right)^n = 0$ 发散

11. 交换 $\int_0^1 dx \int_0^{\sqrt{x}} f(x, y) dy$



$\int_0^1 dy \int_{y^2}^1 f(x, y) dx$

12. 格林公式

$\oint_L y^2 dx + x^2 dy = \iint_D (2y - 2x) dx dy$

13. $\sum_{n=1}^{\infty} x^n$ 收敛半径

$\lim_{n \rightarrow \infty} \frac{x^{n+1}}{x^n} = x < 1 \Rightarrow R=1$

14. $z = f(x, y)$ 求 dz

$z \begin{cases} u = x \\ v = y \end{cases}$

$dz = (f'_1 \cdot 1 + f'_2 \cdot 1)dx + (f'_3 \cdot 1)dy$

15. 已知 α, β, γ 是曲面的一族

$\int \int_S p dx dy + q dy dz + r dz dx$

他换了 p, q, r 对 α, β, γ

$\int \int_S (p \cos \alpha + q \cos \beta + r \cos \gamma) dS$

16. $z = x^3 + y^3 + 3x^2 - 3y^2 - 12x - 12y + 1$ 求极值

$f_x = 6x^2 + 6x - 12 = 0$

$f_y = 6y^2 - 6y - 12 = 0$

$f_{xx} = 12x + 6$
 $f_{yy} = 12y - 6$

$f_{xy} = 0$

$f_{xx} = 12x + 6$

$f_{yy} = 12y - 6$

① $(-2, 2)$ 非

② $(2, -1)$

$0 - 18 \times (-18) < 0$

$A = -18 < 0$ 极大

$f(2, -1) = -16 - 2 + 12 - 3 + 24 + 12 + 1$

③ $(1, -1)$ 非

④ $(1, 2)$ $0 - 18 \times (-18) < 0$

$A = 18 > 0$ 极小

$f(1, 2) = 2 + 8 + 3 - 12 - 12 - 24 + 1$

$f(1, 2) = 2 + 16 + 3 - 12 - 12 - 24 + 1$

$= 22 - 48$

17. $I = \int \int_D xy dx dy$ $D: y^2 = x$ 与 $y = x$ 围成

18. 常数变易法求 $y' + y = e^{-x}$ 通解

19. $f(x) = e^{2x}$ 求 n 阶导数

(1) $f'(x) = 2e^{2x}$

$f''(x) = 2^2 e^{2x}$

$f^{(n)}(x) = 2^n e^{2x}$

(2). $e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots + \frac{x^n}{n!} + \dots$

$\sum_{n=0}^{\infty} \frac{x^n}{n!}$

20. a, b, c 为常数

求方向向上上半椭球面

$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1 (z > 0)$

$I = \int \int_S xy^2 dy dz + yz^2 dz dx + zx^2 dx dy$